



Multi-Source BFS Pattern Sheet

(One of the Most Important Graph/Grid Interview Patterns)

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1. What is BFS?

BFS stands for

Breadth First Search

Unlike DFS,

DFS

↓

Go Deep

BFS explores

Level by Level



Layer by Layer

Example



BFS Order

A



B C D



E F

DFS

A



B



E



Backtrack



F

↓

C

↓

D

2. Why BFS Finds Shortest Distance

Suppose

S . . .

. . . .

. . . D

Each move costs

1

BFS explores

Distance = 0

↓

Distance = 1

↓

Distance = 2

↓

Distance = 3

The first time we reach

Destination

it is guaranteed to be

Minimum Distance

DFS cannot guarantee this.

Visual

Layer 0

S

↓

Layer 1

*

↓

Layer 2

**

↓

Layer 3

↓

Destination

3. When to Use BFS Over DFS

Whenever the problem asks

Nearest

Minimum

Shortest

Distance

Steps

Moves

Minutes

Time

Levels

Spread

Infection

Fire

Rotting

Wave

Closest

Immediately think

BFS

Why NOT DFS?

DFS

Go Deep

↓

May find a very long path first

↓

Not guaranteed shortest

BFS

Explore all shorter paths first

↓

Guaranteed shortest

BFS Recognition Keywords

Nearest

Closest

Minimum Distance

Shortest Path

Minimum Steps

Minimum Moves

Minutes

Hours

Time Taken

Spread

Rot

Wave

Flood

Infection

4. Single Source BFS

Meaning

One Starting Point

Example

```
S . .  
. . .  
. . D
```

Queue initially

```
[S]
```

BFS expands

```
S  
↓  
Neighbours  
↓  
Next Layer  
↓  
Next Layer
```

Generic Single Source BFS Template

```
from collections import deque  
  
q = deque()  
  
visited = [[False]*m for _ in range(n)]  
  
q.append((sr, sc))  
visited[sr][sc] = True  
  
while q:  
    r, c = q.popleft()  
    for dr, dc in dirs:
```

```
nr = r + dr
nc = c + dc

if 0 <= nr < n and 0 <= nc < m:

    if not visited[nr][nc]:

        visited[nr][nc] = True

        q.append((nr, nc))
```

Time Complexity

$O(V + E)$

Grid

↓

$O(n*m)$

5. Multi-Source BFS

Suppose

```
. . . .
. . . .
. . . .
. . . .
```

Question

Distance
to nearest fire

Brute Force


```
Fire 1
↓
Entire BFS

Fire 2
↓
Entire BFS

Fire 3
↓
Entire BFS
```

Huge repeated work.

Instead

Start

```
ALL
fires
together.
```

Queue initially

```
[(0,0),
(2,3),
(3,1)]
```

One BFS.

Why It Works

Imagine

Fire spreads
every minute.

Minute 0



Minute 1

Adjacent

Minute 2

Next Layer

Exactly

what BFS naturally does.

6. Multi-Source BFS Recognition

Ask

Are multiple sources
spreading
simultaneously?

If

YES

Think

Multi-Source BFS

Common Keywords

Nearest Source

Nearest Zero

Nearest Gate

Rotting

Infection

Fire

Spread

Wave

Simultaneously

Minimum Distance

Closest

Multi-Source BFS Idea

Instead of

One Source

↓

BFS

Think

All Sources

↓

ONE BFS

Biggest Difference

Single Source

Queue
↓
One Cell

Multi Source

Queue
↓
ALL Starting Cells

Nothing else changes.

7. BFS vs DFS

Feature	DFS	BFS
Data Structure	Stack / Recursion	Queue
Traversal	Depth First	Level Order
Shortest Distance	✗	✓
Reachability	✓	✓
Connected Components	✓	✓
Distance Problems	✗	✓
Flood Fill	✓	✓
Multi Source	Rare	Very Common

When DFS is Better

Can Reach?
Connected?
Island?
Boundary?
Cycle?
Component?
Backtracking?

When BFS is Better

Nearest?
Shortest?
Minimum?
Distance?
Time?
Minutes?
Spread?

8. Single Source vs Multi-Source BFS

Feature	Single Source	Multi Source
Starting Nodes	One	Many
Queue Initially	One Cell	All Sources
Traversal	Same	Same
Complexity	$O(V+E)$	$O(V+E)$

Feature	Single Source	Multi Source
Typical Problems	Shortest Path	Spread / Nearest Source

9. Generic BFS Template

```
from collections import deque

q = deque()

q.append(start)

visited[start] = True

while q:

    node = q.popleft()

    for neighbour in neighbours:

        if not visited[neighbour]:

            visited[neighbour] = True

            q.append(neighbour)
```

10. Generic Multi-Source BFS Template

```
from collections import deque

q = deque()

visited = [[False]*m for _ in range(n)]

# Push ALL sources

for i in range(n):
    for j in range(m):

        if grid[i][j] is SOURCE:

            q.append((i, j))

            visited[i][j] = True

while q:
```

```
r, c = q.popleft()

for dr, dc in dirs:

    nr = r + dr
    nc = c + dc

    if 0 <= nr < n and 0 <= nc < m:

        if not visited[nr][nc]:

            visited[nr][nc] = True

            q.append((nr, nc))
```

Multi-Source BFS with Distance

```
from collections import deque

q = deque()

dist = [[-1]*m for _ in range(n)]

for every source:

    q.append(source)

    dist[source] = 0

while q:

    r, c = q.popleft()

    for neighbour:

        if dist[neighbour] == -1:

            dist[neighbour] = dist[r][c] + 1

            q.append(neighbour)
```

11. Recognition Checklist

Ask yourself

Is this asking
Shortest Distance?

↓

YES

↓

Does movement cost
1 step?

↓

YES

↓

Are there
multiple starting points?

↓

YES

↓

Multi-Source BFS

Otherwise

Single Source BFS

12. Universal Pattern Template

Problem

↓

Need Minimum Distance?

↓

YES

↓

BFS

↓

Multiple Sources?

↓

YES

↓

Put ALL sources into Queue

↓

Level Order Expansion

↓

Answer

13. Similar Interview Problems

Single Source BFS

- Binary Maze
- Shortest Path in Binary Matrix
- Word Ladder
- Knight Minimum Moves

Multi-Source BFS

- 01 Matrix ★★★★★
- Rotting Oranges ★★★★★
- Walls and Gates ★★★★★
- As Far From Land As Possible ★★★★★
- Shortest Bridge ★★★★★

14. Real World Examples

Virus Spread

Initially

5 infected people

↓

Virus spreads

simultaneously

Multi-Source BFS

Fire Spread

Multiple fire sources

↓

Spread together

Multi-Source BFS

Multiple Hospitals

Nearest Hospital

↓

Distance

Multi-Source BFS

WiFi Routers

Multiple Routers

↓

Signal spreads

Multi-Source BFS

Flood

Water enters

from multiple locations

↓

Spreads together

Multi-Source BFS

15. Ultimate Cheat Sheet

DFS

Reachability

Connected Components

Boundary Problems

Flood Fill

Cycle Detection

BFS

Shortest Distance

Minimum Steps

Nearest

Time

Levels

Multi-Source BFS

Many Sources

↓

One BFS

↓

Nearest Source

↓

Spread

↓

Minimum Distance

Memory Trick

DFS

Can I Reach?

BFS

How Fast Can I Reach?

Multi-Source BFS

Everyone Starts Together

↓

Spread Level by Level

One-Line Interview Rule

If the problem asks "Can I reach?" → think DFS.

If it asks "How many steps/minutes/shortest distance?" → think BFS.

If there are multiple starting points spreading simultaneously, initialize the queue with all of them and use Multi-Source BFS.